

Audit of `body.tex` against the Lean verification

Statement-correspondence report

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Executive conclusion

The principal mathematical claims agree with the Lean development, subject to the formal scope described below. All thirteen labeled theorem or lemma statements in `body.tex` have corresponding proved Lean declarations; no contradictory bound or missing hypothesis was found in those displayed results. The asymptotic notation $\lesssim_k$ is justified by stronger, explicit polynomial bounds in the code.

The body as a whole should nevertheless not be described as literally proved by Lean without qualification. There are two material wording issues:

1. The Lean development starts from an abstract family of real vectors, scalars, and error terms satisfying the `LanczosRun` interface. It does not prove that executing Algorithm 1 in IEEE floating point produces such an interface.
2. Exact preservation of starting spectral weights holds between the original matrix A and the unperturbed dilation S . For the nearby exact Lanczos problem $\tilde{T}$, Lean proves perturbative Wasserstein and separated-cluster mass bounds, not exact equality of starting weights.

Thus the appropriate verdict is: **the labeled mathematical results match; the algorithmic and spectral-weight prose needs qualification.**

Verification performed

The following checks were run directly on the supplied sources.

- `lake build` completed successfully (3,238 jobs). The output contained linter warnings but no build error.
- `lake env lean Audit.lean` checked the principal declarations. Each depends only on `propext`, `Classical.choice`, and `Quot.sound`, the standard logical axioms used by Lean/Mathlib.
- A source search found no `sorry`, `admit`, or user-declared `axiom` in the Lean files.
- The pinned toolchain is Lean 4.32.2.

The authoritative formal input is the structure `FinitePrecisionLanczos.LanczosRun` in `Lanczos/Run.lean`. Its important global hypotheses are

$$H^T = H, \quad \varepsilon \geq 0, \quad k > 0, \quad k\varepsilon \leq \frac{1}{100},$$

together with the five local arithmetic identities and error bounds, near-normalization, nonnegative β_j , and strict positivity of the internal subdiagonals $\beta_1, \dots, \beta_{k-1}$.

Correspondence of labeled results

In the table, “match” means that the body statement follows immediately from the cited Lean declaration after replacing the explicit polynomial bound by $\lesssim_k$. Line numbers refer to the supplied `body.tex`.

Body result	Principal Lean declaration	Finding
Local error bounds (lines 265–275)	<code>LanczosRun.local_errors</code>	Match. Lean proves $\ f_j\ _2 \leq 700j\varepsilon\ H\ _2$, $ g_j \leq \varepsilon$, and $ p_j \leq 1200j\varepsilon\ H\ _2$. The exact identity used for p_j omits e_j^u , confirming the stated cancellation.
A priori coefficient bounds (282–290)	<code>LanczosRun.coefficients</code>	Match. Each of $\ u_j\ _2$, $\ w_j\ _2$, $ \alpha_j $, $\ z_j\ _2$ and β_j is at most $100j\ H\ _2$.
Gram commutator (447–464)	<code>LanczosRun.gram_commutator_conclusion</code>	Match. Lean proves both exact commutator identities, the support statement for the diagonal-defect commutator, and explicit entry bounds.
Paige loss bound (521–529)	<code>LanczosRun.paige_bound</code>	Match. Lean proves the absolute boundary product is at most $6000k^3\varepsilon\ H\ _2$, which is exactly the body quantity after regrouping β_k .
Paige descent (577–586)	<code>LanczosRun.paige_descent</code>	Match. The two conclusions are both bounded by $8t(6000k^3\varepsilon\ H\ _2)$ for a smaller prefix.
Ritz containment (709–716)	<code>LanczosRun.containment_prefix</code> , <code>LanczosRun.containment_eigenvalue</code>	Match. Lean proves the stronger explicit radius $[(8t^2 + 5)6000k^3 + 2100k^2]\varepsilon\ H\ _2$ for every prefix T_t .
Stabilized Ritz localization (784–793)	<code>LanczosRun.stabilized</code> , <code>LanczosRun.terminal_localization_eigenvalue</code>	Match. The code gives $3\beta_k y_k $ plus an explicit $O(k^5\varepsilon\ H\ _2)$ term, and the terminal conclusion when $\beta_k = 0$.
Normalized local errors (850–859)	<code>LanczosRun.normalized_local_errors</code> , <code>LanczosRun.normalized_governing</code>	Match. The residual is at most $2600k^4\varepsilon\ H\ _2$ and both kinds of weighted adjacent product are at most $4800k\varepsilon\ H\ _2$.
Normalized commutator (892–904)	<code>LanczosRun.normalized_gram_commutator</code>	Match. Upper triangularity and the exact skew identity are proved, with norm bound $50000k^5\varepsilon\ H\ _2$.
Nilpotent isometry (950–961)	<code>LanczosRun.nilpotent_isometry</code>	Match. Strict upper triangularity, $K^k = 0$, $\ K\ _2 \leq 1$, $\ C\ _2 \leq 2$, and all three displayed identities are formal theorems.
Physical dilation (1001–1017)	<code>LanczosRun.physical_dilation</code> , <code>LanczosRun.block_recurrence</code>	Match. Lean proves the orthogonality, first-column, recurrence, and norm claims, with $\ \hat{E}\ _2 \leq 10^6k^8\varepsilon\ H\ _2$.

Body result	Principal Lean declaration	Finding
Symmetric completion (1110–1120)	<code>LanczosRun.greenbaum_exact_completion,</code> <code>GreenbaumModel.prescribed_subdiagonal_positive</code>	Match. The leading block is exactly T_k , the completed matrix is symmetric tridiagonal, and the operator difference is at most $3\ \hat{E}\ _2$. Positivity is guaranteed through the leading block.
Greenbaum backward model (1146–1159)	<code>LanczosRun.exists_greenbaumModel,</code> <code>GreenbaumModel.exact_lanczos_returns_Tmat</code>	Qualified. The matrix, dimension, leading block, and $3,000,000k^8\epsilon\ H\ _2$ bound match. Exact reproduction is proved for the formal <code>ExactLanczos</code> interface, not by verification of a particular software implementation.

Scope difference: Algorithm 1 versus `LanczosRun`

The body moves from kernel assumptions and Algorithm 1 to a completed run. The Lean code does not contain floating-point values, a rounding operator `fl`, IEEE 754 semantics, or an implementation of Algorithm 1. Instead, every run-level theorem accepts the local identities and bounds as fields of `LanczosRun`. Consequently, statements such as

“Suppose Algorithm 1 completes k iterations under the assumptions of the model”

are supported only after an external proof supplies the bridge

execution of Algorithm 1 $\implies$ an instance of `LanczosRun`.

No such bridge is present in the supplied Lean project. This does not weaken the conditional matrix analysis; it limits what has been verified about an actual floating-point program.

There are also two benign differences showing that the formal interface is slightly more general than the paper’s operational description. Lean does not assume $k \leq n$, and when $\beta_k = 0$ it does not require separately that $z_k = \Delta_k = 0$; it needs only the stated local recurrence identity and error bound. Every body run satisfying its conventions is still covered.

Material wording issue: starting spectral weights

The prose at lines 819 and 1261 says that the construction or model “preserves the starting spectral weights exactly.” This is correct only for the unperturbed dilation S , which is orthogonally similar to k copies of A . Lean proves

$$\mu_{S,e_1} = \mu_{A,\bar{v}_1}$$

in `GreenbaumModel.starting_spectralMass` (and proves equality of every starting moment in `starting_moment`).

The nearby problem on which exact Lanczos reproduces T_k is instead $\tilde{T}$. Between A and $\tilde{T}$, Lean proves

$$\mathcal{W}_1^{\text{dual}}(\mu_{A,\bar{v}_1}, \mu_{\tilde{T},e_1}) \leq \sqrt{nk} \delta, \quad \delta = \|S - \tilde{T}\|_2 \leq 3,000,000k^8\epsilon\|H\|_2,$$

not exact equality. It also proves a cluster-mass estimate only when the chosen physical cluster is separated by a gap $g > 2\delta$:

$$|\text{physical mass} - \text{model cluster mass}| \leq \frac{\sqrt{nk} \delta}{g - 2\delta}.$$

A precise replacement for the disputed wording is:

“The unperturbed dilation preserves the starting spectral measure exactly; the nearby completed problem differs from it by one operator perturbation, which yields dual-Wasserstein and separated-cluster mass bounds.”

With this change, the spectral interpretation matches the formal results.

Claims outside the Lean verification

The following parts of the body are not contradicted by the code, but they are not established by its declarations and should not be presented as machine-checked consequences:

- the IEEE single-precision experiments and all numerical values in the three figure discussions;
- historical and literature-priority claims;
- broad empirical statements about practical convergence and accuracy of Lanczos-based matrix-function approximations;
- the informal eigenvector-accuracy remark following the Ritz residual bound (standard perturbation reasoning, but not packaged in this Lean development).

The Lean development actually proves two useful spectral consequences that the main backward theorem in the body does not state explicitly: every eigenvalue of $\tilde{T}$ lies within δ of $\text{spec}(H)$, and the starting measures satisfy the dual-Wasserstein and separated-cluster bounds displayed above.

Recommended minimal edits to `body.tex`

1. After the sentence about the companion Lean proof (line 93), add: “The formalization begins from the local real-arithmetic run interface displayed below; it does not formalize a floating-point implementation of Algorithm 1.”
2. At lines 819 and 1261, replace unqualified exact preservation of “the model” weights by the precise S versus $\tilde{T}$ statement given above.
3. In the consequence of Theorem `thm:greenbaum`, describe exact Lanczos as the formal exact three-term-recurrence interface, or define that interface explicitly.

After those edits, the paper-facing theorem statements and the formal scope would be aligned without a material caveat.